

Write Efficiency Improvement and Potential of SOT-MRAMs

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MTJs for SOT-MRAM were optimized by varying the thickness of the storage-layer. Scalabilities of the write-efficiency and critical switching-current was found to be excellent for lateral size of 65-150nm. Exploration of the retention-energy

Finally, possibility of reducing I_{csw} and improving the write-efficiency were estimate by using the equation. The estimation shows the good possibility to break the values for conventional STT-MRAMs.

Index Terms—MRAM, SOT, Strain, Write-efficiency

I. INTRODUCTION

Advances in spintronics—most notably room-temperature tunneling magnetoresistance and spin torque switching—have enabled the commercialization of both field writing MRAM and spin transfer torque (STT) MRAM. To date, however, their use has been largely limited to relatively low density data storage applications. Consequently, the overall market size of MRAM technologies remains modest compared with that of the broader semiconductor industry.

The next major target for MRAM may be working memory applications, where virtually unlimited endurance and low write-energy consumption are essential requirements. However, entering this domain is challenging for conventional MRAMs, as their limited endurance and relatively high write-energy consumption continue to pose significant obstacles.

Building on the potential of spintronics, additional switching mechanisms—such as spin orbit torque (SOT) magnetic switching—have recently been demonstrated in MRAM storage cells [1]–[2]. SOT MRAM is considered a particularly promising candidate for future working memory applications because it intrinsically offers unlimited endurance.

Recently, high write-efficiency has been reported using MTJs with strain induced magnetic anisotropy (SIMA MTJs) [2]. If the switching current maintains scalability while ensuring sufficient retention energy for small MTJs with widths of 20–30 nm, SOT MRAM employing SIMA MTJs would represent a highly promising candidate for nonvolatile working memory.

II. MTJ MATERIALS AND STRUCTURE OF MTJs WITH SIMA

Figure 1(a) and (b) show the schematic illustration and MTJ materials used in this study.

The materials and layer structure are almost identical to those of conventional MTJs with in-plane magnetic anisotropy. A Ta-based spin-Hall electrode (SHE) with a spin-Hall angle of approximately 0.10–0.15 and in-plane MTJ were deposited. The storage layer consists of CoFeB, which has a magnetostriction constant of approximately 4×10^{-5} . The MTJ stack was patterned into a rectangular shape with an aspect ratio of 1:1.5 using ion-beam etching. Anisotropic strain was

introduced to induce SIMA in the MTJs, following the same concept reported previously [1].

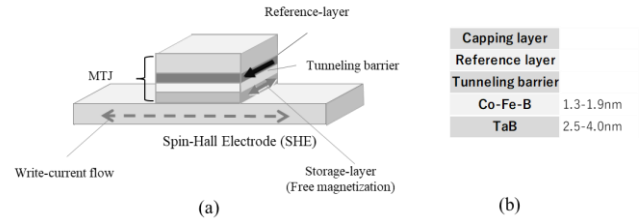


Fig. 1. Schematic illustration(a) of SIMA-MTJ and MTJ materials(b).

III. STORAGE-LAYER THICKNESS OPTIMIZATION FOR MAXIMIZING WRITE-EFFICIENCY OF SOT-WRITING

The storage-layer takes perpendicular magnetization below 1.3nm thickness and does in-plane magnetization above 1.3nm. Therefore, the thickness, t_{SL} , was varied between 1.3nm and 1.9nm in order to maximize write-efficiency, $\Delta E_r / I_{\text{csw}}$. Here, I_{csw} is critical switching-current with write-current pulse-width of 10ns and ΔE_r is retention-energy.

Figures 2(a), 2(b), and 2(c) show the t_{SL} dependences of I_{csw} , ΔE_r , and $\Delta E_r / I_{\text{csw}}$, respectively.

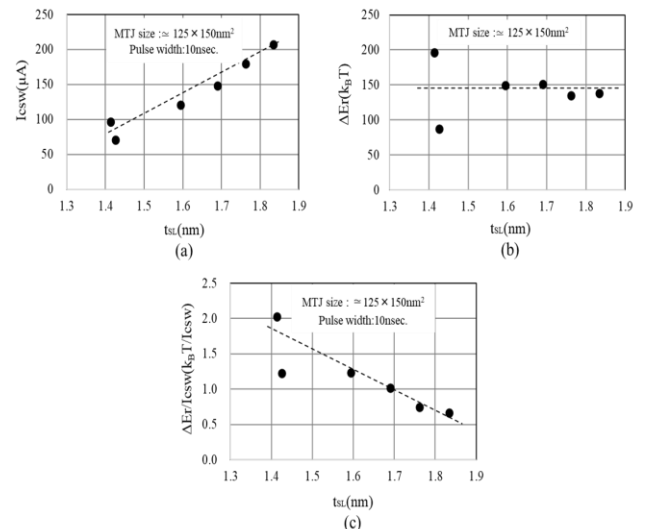


Fig. 2 t_{SL} dependences of I_{csw} (a), ΔE_r (b), and $\Delta E_r / I_{\text{csw}}$ (c) takes maximum at t_{SL} of about 1.4–1.5nm.

Retention energy ΔE_r was estimated by fitting the switching probability data obtained from repeated resistance-vs-applied magnetic field measurement with the Sharrock's formula[3].

ΔE_r seemed very sensitive to the etching-process and there is a large variation. However, as t_{SL} decreases, I_{CSW} decreases more rapidly than ΔE_r does.

As a result, $\Delta E_r/I_{CSW}$ reached its maximum at $t_{SL} \approx 1.4$ – 1.5 nm.

IV. EXCELLENT WRITE-EFFICIENCY SCALABILITY AND CRITICAL SWITCHING-CURRENT SCALABILITY

I_{CSW} and ΔE_r were evaluated for SIMA MTJs with aspect-ratio of about 1-1.5.

Figure 3(a) shows the MTJ-length dependence of write-efficiency. All the write-efficiencies in the Fig.3 were evaluated for MTJs with $t_{SL} \approx 1.5$ nm. Smaller MTJs exhibit improved write-efficiency and a very high efficiency of $4.1 \text{ k}_B\text{T}/\mu\text{A}$ was obtained by the smallest MTJ with $65\text{nm} \times 65\text{nm}$ in lateral size. MTJs with small aspect-ratio seems to have better write-efficiency than those with large aspect-ratio. This is one of the merits of SIMA-MTJs.

Figure 3(b) shows the MTJ length dependence of I_{CSW} . Regardless of thickness range of 1.41 – 1.61 nm, I_{CSW} decreases as the MTJ length becomes smaller. This reduction directly contributes to the excellent write-efficiency scalability.

If the retention energy remains sufficient even for small MTJs, SIMA-MTJs can provide scalable performance as nonvolatile working memory.

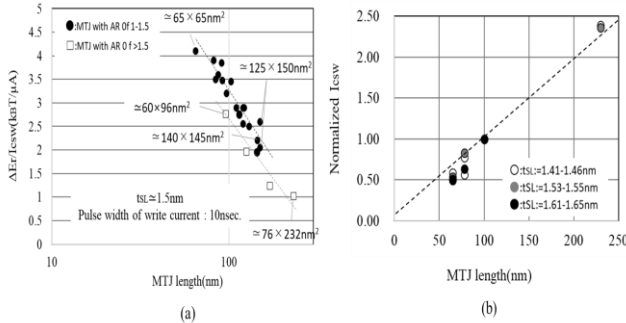


Fig. 3. MTJ length dependences of $\Delta E_r/I_{CSW}$ (a) and Normalized I_{CSW} . Due to the excellent I_{CSW} scalability, $\Delta E_r/I_{CSW}$ shows good scalability.

V. RETENTION ENERGY FOR SMALL MTJs

Figure 4 shows the width dependence of retention-energy for SIMA-MTJs with aspect-ratio of 1-1.5. The storage-layer was 1.6 nm thick.

Even with the small aspect-ratio, SIMA-MTJs have large retention-energy over $200 \text{ k}_B\text{T}$.

The dotted extrapolation-lines indicate that even for MTJ width of 20 – 30 nm , the retention energy can remain as high as 50 – $120 \text{ k}_B\text{T}$, suggesting that sufficient nonvolatility can be maintained at such small dimensions.

From this study, it is concluded that SOT-MRAM using SIMA-MTJs demonstrates their potential as nonvolatile working memory.

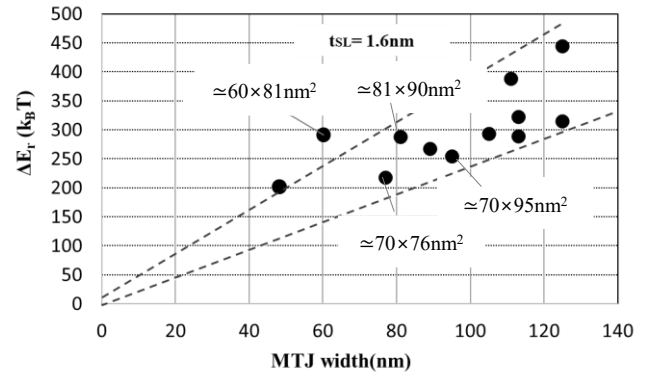


Fig.4 MTJ width dependence of ΔE_r . The dotted extrapolation-lines indicate that even for MTJ width of 20 – 30 nm , the retention energy ΔE_r can remain as high as 50 – $120 \text{ k}_B\text{T}$.

VI. SUMMARY

The critical switching current and the retention energy were evaluated for SIMA-MTJs with lateral dimensions of approximately 65 – 150 nm . The write efficiency was found to be highly scalable, primarily because I_{CSW} itself exhibits excellent scalability. Furthermore, the width dependence of ΔE_r indicates that SIMA-MTJs with widths of 20 – 30 nm would still maintain sufficient retention-energy for 10 year data-retention, demonstrating their potential as nonvolatile working memory.

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